

diffusion MRI (dMRI) acquisition, which mitigates crossing fiber limitations inherent in the more traditional diffusion tensor imaging (DTI) model and enables the use of biophysical diffusion models, such as the neurite orientation dispersion and density imaging (NODDI) model, to assess more specific tissue microstructural properties.¹³ Given the evidence suggesting that the LC-EC tract is an early predilection pathway contributing to potential proteinopathic transmission processes during the earliest stages of AD, examining LC-EC microstructural integrity in the context of early AD pathology could provide further insights into in vivo detectable pathological events preceding preclinical AD.

Guided by the hypothesis that elevated levels of AD pathology may be associated with fine-grained microstructural changes in the LC-EC, we sought to investigate the microstructural integrity of the pathway using 7T multishell dMRI data and assessed its relation to several plasma AD-related biomarkers in cognitively unimpaired individuals across the adult lifespan.

2 | MATERIALS AND METHODS

2.1 | Participants

A subgroup of cognitively unimpaired individuals across the adult lifespan ($n = 47$; mean age = 60.4 ± 15.1 years; age range = 30 to 85; 23 females [48.94%]) from the 7T Adult Lifespan Cohort^{14–16} was included in the present study based on the availability of diffusion-weighted imaging data. Participants were recruited for the original study from the general population in the southernmost region of the Netherlands. Exclusion criteria included ultra-high-field imaging contraindications, a history of major psychiatric or neurological disorders, history of brain injury or brain surgery, history of major vascular disorders, use of drugs or psychoactive medication, or excessive alcohol consumption (>15 units/week).

2.2 | 7T MRI data acquisition

All participants underwent an imaging protocol on a 7T MAGNETOM Siemens scanner (Siemens Healthineers, Erlangen, Germany) with a 32-channel (1TX/32RX) head coil (Nova Medical, Wilmington, MA, USA). First, a whole-brain magnetization-prepared 2 rapid acquisition gradient-echo (MP2RAGE) sequence¹⁷ (TR = 5000 ms, TE = 2.47 ms, flip angle = $5^\circ/3^\circ$, voxel size = 0.7 mm isotropic, number of slices = 240) was acquired. The LC was imaged using an in-house developed magnetization transfer-weighted turbo flash (MT-TFL) sequence¹² multishot 3D readout, repetition time (TR) = 538 ms, echo time (TE) = 4.08, flip angle = 8° , voxel size = $0.4 \times 0.4 \times 0.5$ mm, number of slices = 60, with center-out k-space sampling and preceded by 20 long off-resonant Gaussian sinc pulses (pulse length = 5.12 ms, bandwidth = 250 Hz, $B_1 = 0.25 \mu\text{T}$). The field of view of the MT-TFL sequence was placed perpendicular to the dorsal surface of the pons and encompassed the area between the inferior colliculi and the caudal border of the pons. A multishell high-angular-resolution DWI sequence (66 directions, $b = 2000$

RESEARCH IN CONTEXT

- 1. Systematic review:** A literature review using traditional sources (e.g., PubMed) indicated that integrity of the LC-EC tract is altered in the context of AD-related pathophysiological processes. In vivo neuroimaging evidence relating LC-EC integrity to AD pathologic processes in preclinical stages of the disease remains limited. Investigating the associations between fine-grained changes in the LC-EC pathway, and early AD pathology may facilitate the identification of individuals at greater risk for AD.
- 2. Interpretation:** Worse 7T MRI-derived LC-EC tract microstructural integrity metrics are associated with elevated levels of plasma phosphorylated tau and GFAP in cognitively unimpaired adults. These associations show topographical specificity, particularly around bending regions of the tract.
- 3. Future directions:** Longitudinal studies using repeated assessment of LC-EC pathway integrity and AD biomarkers will be critical to better understand the pathological mechanisms underlying the observed relationships and to elucidate the temporal ordering of the events at play in asymptomatic individuals.

s/mm^2 ; 35 directions, $b = 700 \text{ s/mm}^2$; six directions, $b = 20 \text{ s/mm}^2$; $b = 0$ in opposite phase-encoding direction; TR = 5000 ms, TE = 60.8 ms, voxel size = 1.25 mm isotropic, number of slices = 72) was used to acquire diffusion data. Given our focus on the brainstem, the field of view of the DWI acquisitions omitted the dorsal-most part of the frontal/parietal lobes.

2.3 | Structural MRI data preprocessing

Whole-brain MP2RAGE images were processed using FreeSurfer¹⁸ (version 6.0.0; <https://surfer.nmr.mgh.harvard.edu>) with the software's default protocol for cortical and subcortical reconstruction, as described in Dale et al.,¹⁹ including the expert options for 7T data. Shortly, each T1-weighted image underwent an automated segmentation process including intensity normalization, skull stripping, segregation of the left and right hemispheres, removal of the brainstem and cerebellar areas, correction of topology defects, definition of the borders between gray matter (GM) and WM and between GM and cerebrospinal fluid (CSF), and parcellation of cortical and subcortical areas. Subsequently, all images underwent visual inspection and, if necessary, manual editing. Finally, the parcellated areas were registered to the MP2RAGE space using the function Estimate and Reslice with the nearest neighbor interpolation option in SPM12 software (<https://www.fil.ion.ucl.ac.uk/spm/software/spm12/>).

A detailed description of our 7T LC preprocessing pipeline can be found in Van Egroo et al.^{15,16} In summary, individual MT-TFL scans were intensity-normalized based on a subject-specific mean intensity of a 10×10 -voxel region of interest (ROI) located in the pontine tegmentum (PT). Next, a study-specific MT-TFL template was created using an iterative diffeomorphic warp estimate in the Advanced Normalization Tools (ANTs) software. The LC mask was then manually delineated on the resulting template by an expert (MVE), based on voxel intensities and prior knowledge of the anatomical properties of the LC.

2.4 | Diffusion MRI data preprocessing

Diffusion-weighted scans were corrected for B_0 susceptibility-induced and eddy current-induced distortions and motion artifacts employing topup and eddy tools of the FSL software²⁰ (version 5.0.9). Whole-brain maps of DTI-based markers of tissue microstructure such as fractional anisotropy (FA) and mean diffusivity (MD) were computed using FSL's *dtifit* (version 6.0.7.6) on the preprocessed DWI data. NODDI-derived parameter maps of the neurite density index (NDI) and orientation dispersion index (ODI) were obtained using the Microstructure Diffusion Toolbox (version 1.2.6, <https://github.com/robbert-harms/MDT>). By adopting a tissue model capable of distinguishing three types of microstructural environment, namely, the intracellular, the extracellular, and the free water compartment, the NODDI model allows for a more specific characterization of sub-voxel tissue features compared to DTI-based metrics.¹³ In particular, NDI values range from 0 to 1, with lower values indicating lower neurite density (i.e., predominantly extracellular diffusion) and higher values indicating high neurite density (i.e., predominantly intracellular diffusion). ODI values range from 0 to 1, with lower values reflecting lower neurite arborization/angular variation (i.e., parallel neurites) and higher values reflecting higher neurite arborization/angular variation (i.e., highly dispersed neurites). Moreover, NODDI can better distinguish different neurobiological processes that may confound DTI metrics at a voxel level: for instance, a reduction in FA may be caused by an underlying reduction in NDI, an increased ODI, or other microstructural changes in the tissue.²¹ By leveraging our high-angular, multishell diffusion imaging protocol, the NODDI model allows for the disentanglement and separate analysis of these two key factors contributing to FA.

2.5 | ROI definition and tractography

The LC mask was first brought back from the study-specific MT-TFL template space to the individual structural space using the inverse of the transformations previously calculated by ANTs. The EC ROI was extracted from the FreeSurfer parcellation as per the Desikan–Killiany atlas,²² while the more fine-grained parcellation provided by the Destrieux atlas was used for the creation of the composite (pre-)frontal areas to be used in the control analysis (Supplementary

Methods). Then, individual MP2RAGE images were coregistered to the individual mean diffusion-weighted b_0 volume using the SPM12 Estimate and Reslice function with the nearest neighbor interpolation method, and the resulting transformation matrices were applied to the individual LC masks, EC, and (pre-)frontal parcellations to obtain the LC, EC, and (pre-)frontal ROIs into the individual DWI space for subsequent tractography.

The LC-EC tract of interest was obtained, for each subject, by means of unguided probabilistic tractography using FSL's *bedpostx* (a Bayesian estimation of diffusion parameters allowing for crossing-fiber modeling) and *probtrackx* (Figure 1). For *bedpostx*, the number of fiber populations modeled per voxel was increased to 3 from the default value of 2 to better capture the anatomical complexity of the surveyed region, while other options were left at their default values. In *probtrackx*, the LC was set as a seed ROI, while the EC was set as both waypoint and terminative ROI. To obtain the control tract, the (pre-)frontal composite was used as both waypoint and terminative ROI. Additional computing options were left at their default values, except for the curvature threshold, which was decreased from 0.2 (corresponding to a minimum angle of approximately $\pm 80^\circ$) to 0.0 to allow for the detection of especially sharp bending points, which characterize the LC-EC tract. The tractography was run separately for each hemisphere. The resulting tracts were binarized and visually inspected to ensure that the tractography outcome was not excessively sparse, uneven, or fragmented. For each subject, the LC-EC tract was further pruned by multiplying a binarized (threshold = 0.05) and dilated (kernel operated by a factor of 3) version of the probabilistic atlas computed by Sun et al.,⁹ coregistered to the individual diffusion space. Finally, to avoid redundancies and only examine the WM bundles, we subtracted the LC and EC ROIs from the LC-EC tract and the LC and (pre-)frontal composite ROIs from the control tract.

2.6 | DTI-based and NODDI parameter extraction

FA and MD values computed in *dtifit* and NODDI-derived NDI and ODI metrics were extracted by taking the mean of values across the LC-EC tract. Mean diffusion metrics were averaged across left and right hemispheres, and further statistical analyses were conducted on bilaterally sourced values, as we were not operating under any a priori hypothesis on hemispheric lateralization. The same approach was used to extract DTI and NODDI values in the (pre-)frontal control tract.

2.7 | Slice extraction

To further evaluate any topographical pattern along the LC-EC pathway, we proceeded to extract two-dimensional slices from the individual tracts. Given the shape of the pathway containing points of sharp curvature, we split it into three sections: the caudal-most part, connecting the LC to the thalamic region, was sliced along the y -dimension (left hemisphere: $y = -44$ to -14 , right hemisphere: $y = -44$ to -13);

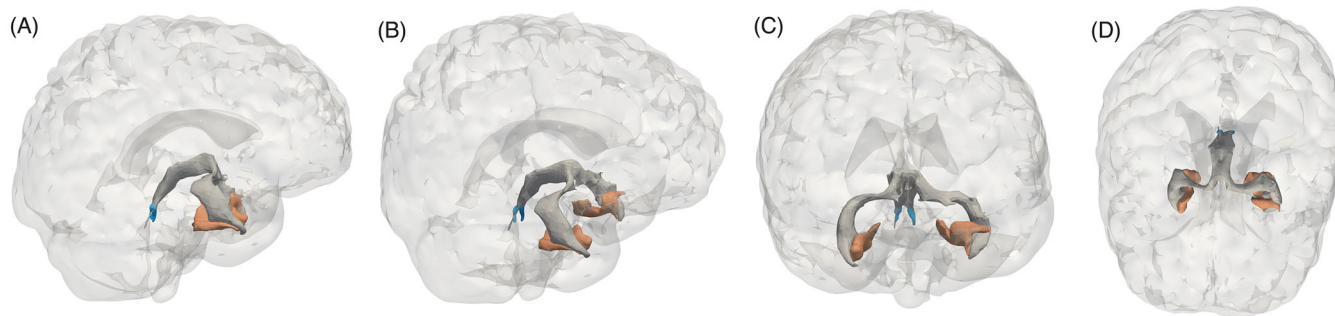


FIGURE 1 LC-EC tract in one representative participant in subject-specific anatomical space. The tract is shown alongside the LC (in blue) and the EC (in orange) in sagittal (A), sagitto-coronal (B), coronal (C), and axial (D) views. EC, entorhinal cortex; LC, locus coeruleus.

the intermediate portion, proceeding laterally toward the temporal lobes, was sliced along the x-dimension (left hemisphere: $x = 2$ to -41 , right hemisphere: $x = -4$ to 40); the inferior portion, terminating in the EC, was sliced along the z-dimension (left hemisphere: $z = -20$ to -48 , right hemisphere: $z = -19$ to -47). To ensure consistency in these nodes across participants, we used the binarized version of the Sun atlas of the LC-EC tract⁹ by extracting, in the individual space, the overlap between the three sections obtained from the atlas and our individual tracts. To accommodate for anatomical variability across individual tracts in native space and the atlas, we used the dilated version (kernel operated by a factor of 3) of the atlas: this procedure explains the slight discrepancy in LC-EC section coordinates between our work and similar efforts.⁷

2.8 | Plasma AD-related assessments

Fasted EDTA plasma samples were collected through venipuncture in the morning and handled according to the standard operating procedure stipulated by the central biobank of Maastricht University Medical Center. Samples were centrifuged at $2000 \times g$, aliquoted in polypropylene tubes, and stored at -80°C in the biobank within 60 min of collection. Plasma biomarkers were analyzed in randomized order using ultra-sensitive Single molecule array (Simoa) assays (Quanterix, Inc.) for amyloid beta ($A\beta$) 42 and $A\beta$ 40 (to create $A\beta$ 42/40 ratio), phosphorylated tau (p-tau) 181 (p-tau181 V2 Advantage Kit), p-tau231 (University Gothenburg),²³ GFAP (Discovery kits for HD-X), and neurofilament light (NfL) (NF-light Advantage Kit) at the University of Gothenburg (Sweden). Analyses were performed in duplicates using a 1:4 automated dilution protocol for all markers, except for 1:2 dilution protocol for p-tau231. Analysis of plasma p-tau217 was performed at Lund University (Sweden) using the Meso Scale Discovery (MSD) platform as previously described.²⁴ Apolipoprotein E (APOE) genotyping was performed using polymerase chain reaction on deoxyribonucleic acid extracted from blood samples, and a participant's APOE status was defined as "ε4 carrier" if they carried at least one ε4 allele. Technicians handling the blood samples were blinded to participant imaging data,

and staff collecting imaging data were blinded to blood assessment results.

2.9 | Statistical analyses

All analyses were performed using R statistical software (version 4.4.1, <https://www.r-project.org/>). As an initial data check, we performed zero-order correlations across all relevant variables with Pearson's correlations and then used robust multiple linear regression models (using the `rlm` function in the MASS package [version 7.3-61], fitted using iterated reweighted least squares implementing a Huber M-estimator; the corresponding p values were extracted using the "frobtest" function in the `sfsmisc` package [version 1.1-19]) to account for potential outliers in biomarkers. Following our main hypothesis, we examined the relationship between DTI-derived (namely, FA and MD) or NODDI-derived (namely, NDI and ODI) metrics in the LC-EC tract and plasma AD-related biomarkers (NfL, GFAP, $A\beta$ 42/40 ratio, p-tau181, p-tau217, and p-tau231). All models were corrected for age, sex, and tract volume and adjusted for multiple comparisons. Sensitivity analyses were conducted by repeating the models in a subsample of older (≥ 50 years) individuals ($N = 35$) and by adding APOE status as a covariate. To determine the specificity of our results, we examined similar associations on the control tract in the entire sample. In a second step, we performed slice-wise associations to explore the topographic distribution. The slices were pooled per three consecutive slices, resulting in 32 bilateral bins, and the associations described were evaluated within each bin.

3 | RESULTS

3.1 | Participant characteristics

Demographic characteristics of the study sample are reported in Table 1. The mean age of the participants was 60 years (range, 30 to 85 years), 23 were females (49%) and 11 (23%) were carriers of at least one APOE ε4 allele.

TABLE 1 Summary statistics for demographic characteristics, diffusion metrics and plasma biomarker levels in the study sample.

N = 47	
Demographic characteristics	
Age, years	60.43 ± 15.11 (30 to 85)
Sex, no. (%)	
Female	23 (48.94)
Male	24 (51.06)
APOE ε4 carrier, no. (%)	11 (23.40)
Volume LC-EC bilateral (mm ³)	4490.46 ± 1775.49 (1364.52 to 10,552.68)
Volume control tract bilateral (mm ³)	65,753.31 ± 17393.89 (31,364.53–105,552.07)
Diffusion metrics	
LC-EC tract (bilateral)	
FA	0.38 ± 0.02 (0.34 to 0.44)
MD (10 ³)	0.64 ± 0.026 (0.59 to 0.71)
NDI	0.59 ± 0.04 (0.49 to 0.67)
ODI	0.23 ± 0.02 (0.18 to 0.27)
Control tract (bilateral)	
FA	0.45 ± 0.02 (0.38 to 0.51)
MD (10 ³)	0.58 ± 0.03 (0.53 to 0.67)
NDI	0.65 ± 0.05 (0.51 to 0.72)
ODI	0.22 ± 0.02 (0.15 to 0.25)
Plasma biomarkers	
GFAP (pg/mL)	153 ± 63.8 (53.8 to 389)
NfL (pg/mL)	18.1 ± 7.68 (6.56 to 43.4)
Aβ42/40 ratio	0.09 ± 0.01 (0.05 to 0.12)
p-tau181 (pg/mL)	1.68 ± 0.86 (0.86 to 6.44)
p-tau217 (pg/mL)	0.27 ± 0.13 (0.11 to 0.81)
p-tau231 (pg/mL)	8.34 ± 3.87 (3.29 to 25.8)

Note: Summary statistics for demographic characteristics, diffusion metrics, and plasma biomarker levels in the study sample. Values are presented as mean ± SD (range) for continuous variables and percentages for dichotomous variables.

Abbreviations: Aβ, amyloid beta; APOE ε4, apolipoprotein ε4; EC, entorhinal cortex; FA, fractional anisotropy; GFAP, glial fibrillary acidic protein; LC, locus coeruleus; MD, mean diffusivity (10³); NDI, neurite density index; NfL, neurofilament light polypeptide; ODI, orientation dispersion index.

3.2 | Correlations of demographics with plasma AD-related biomarkers and LC-EC tract diffusion metrics

Older individuals exhibited lower Aβ42/40 ($t = -3.54$, $p < 0.001$) and higher GFAP ($t = 4.92$, $p < 0.001$), NfL ($t = 7.07$, $p < 0.001$), and p-tau181 ($t = 2.36$, $p = 0.02$) levels (Table S1). Females displayed higher MD ($t = -2.68$, $p = 0.01$) within the LC-EC tract. Older age was associated with lower ODI ($t = -2.91$, $p = 0.006$) in the LC-EC tract. Age and

sex differences remained significant when further controlling for APOE ε4 status

3.3 | Whole-tract relationships between LC-EC tract diffusion metrics and plasma AD-related biomarkers

Lower FA in the LC-EC pathway was associated with higher levels of GFAP ($t = -2.24$, $p = 0.03$, false discovery rate (FDR)-adjusted p value [p_{FDR}] = 0.06) and p-tau217 ($t = -2.07$, $p = 0.04$, $p_{FDR} = 0.09$). Conversely, MD was positively associated with GFAP ($t = 3.81$, $p < 0.001$, $p_{FDR} = 0.002$), p-tau181 ($t = 2.40$, $p = 0.02$, $p_{FDR} = 0.08$), and p-tau217 ($t = 2.21$, $p = 0.03$, $p_{FDR} = 0.09$). Higher ODI in the LC-EC tract was related to elevated p-tau181 ($t = 2.09$, $p = 0.04$, $p_{FDR} = 0.08$) and p-tau231 ($t = 2.49$, $p = 0.02$, $p_{FDR} = 0.07$) (Figure 2, Table 2). No significant relationships were found with NfL values, Aβ42/40 ratio, or NDI values in the LC-EC tract. To examine the specificity of these findings to the LC-EC tract, all analyses were repeated in the control tract, where we observed that a lower Aβ42/40 ratio was associated with lower FA ($t = 3.32$, $p = 0.003$, $p_{FDR} = 0.06$), lower NDI ($t = 2.19$, $p = 0.04$, $p_{FDR} = 0.05$), and higher MD ($t = -3.20$, $p = 0.003$, $p_{FDR} = 0.006$) values, but we detected no significant associations with the other plasma markers (Figure S1; Table S2).

3.4 | Slice-wise relationships between LC-EC tract diffusion metrics and plasma AD-related biomarkers

Taking a more granular approach by using a slice-wise approach along the LC-EC pathway, we observed that the associations involving GFAP, p-tau217, and p-tau231 were driven by the portion of the tract immediately following the medial bend (Figure 3, first vertical dotted line), right after exiting the thalamic area and proceeding towards the temporal lobes, the region around the curvature point leading the pathway to its inferior portion (Figure 3, second vertical dotted line) and the final part of the tract, terminating in the EC. MD associations were mostly observed in the final portion of the tract, close to the EC, whereas the associations involving FA and ODI were more pronounced in the starting region of the tract, from the LC toward the thalamic region.

4 | DISCUSSION

The LC is one of the earliest sites of accumulation of tau pathology, and has been proposed to play a crucial role in the propagation of tau to the first affected cortical site – the EC – via neuron-to-neuron transmission.^{25,26} This seeding process is thought to emerge decades before the onset of AD clinical symptoms.² While supported by *post mortem*^{6,27} and animal^{28,29} studies, in vivo evidence in humans remains limited due to the complexities in accurately identifying the LC and its sharp-bending WM bundle to the EC. Here, by combining recent advances in plasma AD-related biomarkers with dedicated LC

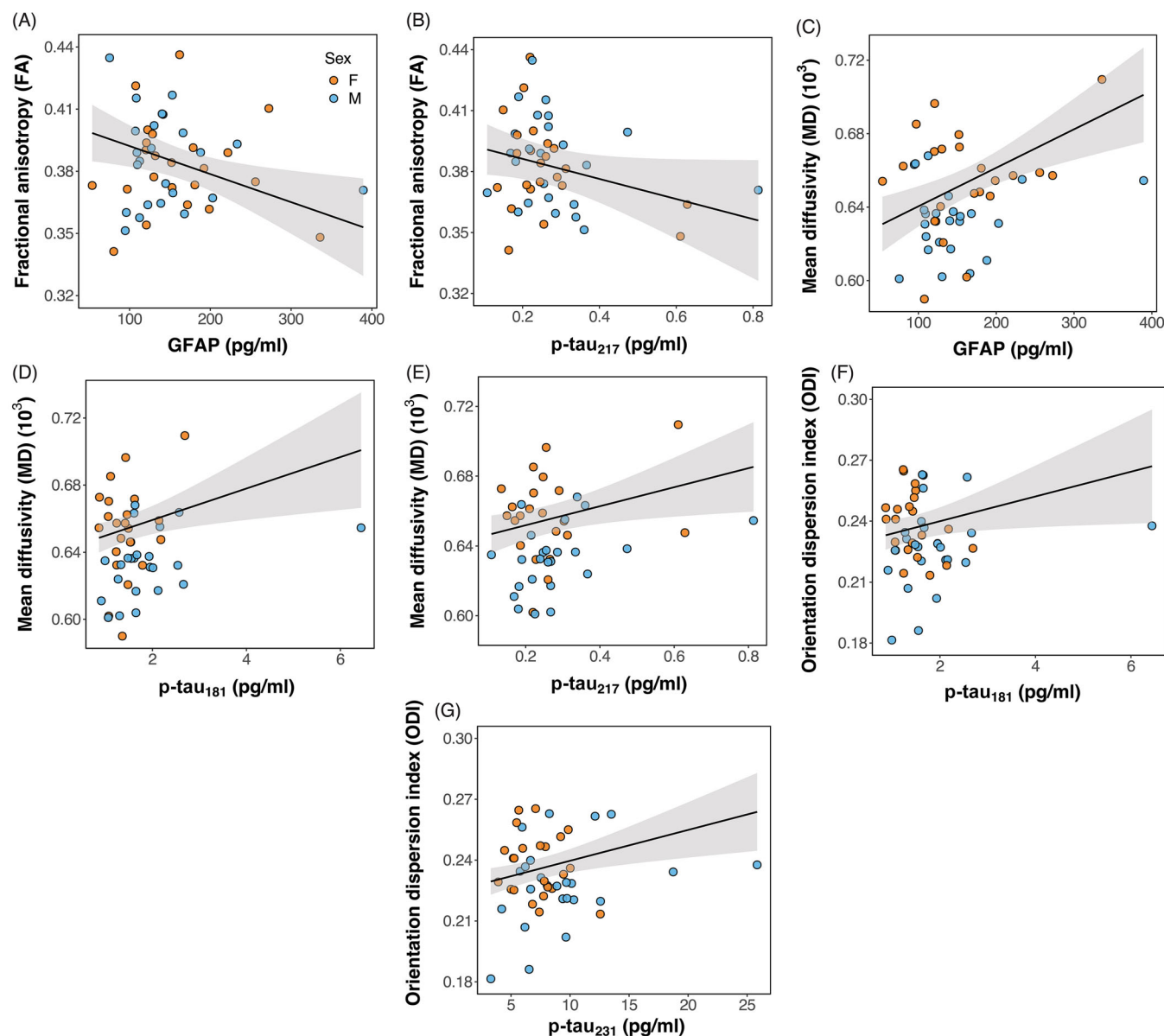


FIGURE 2 Associations between plasma Alzheimer's disease (AD)-related biomarkers ($N = 47$) and LC-EC tract integrity metrics of FA (A, B), MD (C-E) and ODI (F, G). Females are indicated in orange, males in blue. Shaded areas surrounding the regression lines represent the 95% confidence interval of the regression fit. See Table 2 for all the regression coefficients. EC, entorhinal cortex; FA, fractional anisotropy; GFAP, glial fibrillary acidic protein; LC, locus coeruleus; MD, mean diffusivity; ODI, orientation dispersion index.

imaging as well as conventional (DTI) and advanced (NODDI) diffusion models at ultra-high field in vivo, we showed that lower LC-EC tract microstructural integrity was associated with hyperphosphorylated tau burden and astrogliosis in cognitively unimpaired individuals. These results emphasize the potential of microstructural changes within the LC-EC tract as a critical indicator of the earliest AD-related pathophysiological events.

First, we observed that worse LC-EC tract integrity, indicated by lower FA and higher MD values, was related to elevated concentrations of plasma p-tau181 and p-tau217. Similarly, higher ODI within the LC-EC tract was associated with elevated plasma p-tau181 and p-tau231 levels. While p-tau181 is well established for the detection of

existing AD pathology,^{30,31} p-tau217 has shown promise as a sensitive marker for earlier detection of AD,³² potentially offering increased accuracy in detecting AD pathology and higher predictive value for predicting disease trajectory.^{33,34} On the other hand, the p-tau231 epitope reflects a very early detectable process preceding preclinical AD, signaling tau features that precede the formation of neurofibrillary tangles.³⁵ Furthermore, recent work demonstrated the ability of p-tau231 to distinguish positron emission tomography (PET)-derived Braak stage 0 from Braak stages I and II, whereas p-tau181 fails to do so.²³ Our findings, obtained in a cognitively unimpaired cohort, expand on prior reports of lower WM integrity within the LC-EC tract correlating with more advanced clinical AD stages using other diffusion-based

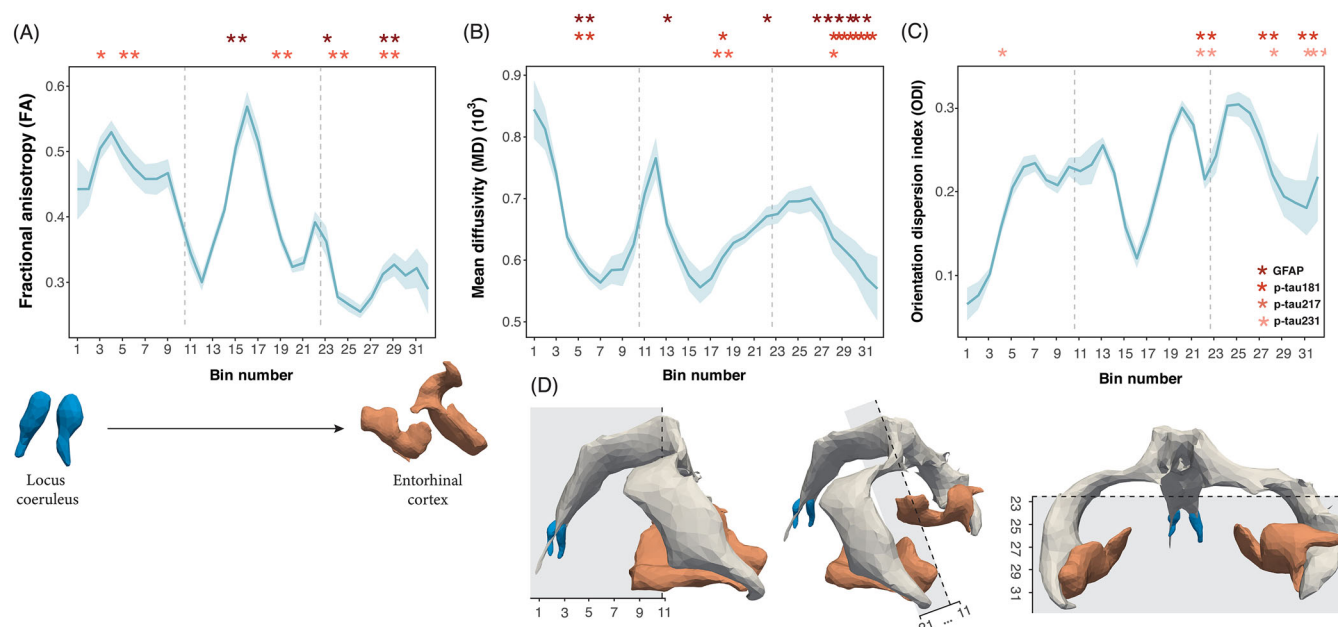


FIGURE 3 (A) FA, (B) MD, and (C) ODI values across LC-EC tract and loci of significance in associations described in main analysis, namely, with GFAP, p-tau181, p-tau217, and p-tau231. The integrity metrics are calculated as mean values across all subjects, with the shaded area representing the 95% confidence interval. Bins result from pooling the slices per three consecutive slices. The vertical dashed lines (A, B, and C) represent the points along the LC-EC tract in which the direction of slicing changed due to a change in the direction of the pathway (D). Statistical significance is indicated as $*p < 0.05$, $**p < 0.01$, $***p < 0.001$. EC, entorhinal cortex; FA, fractional anisotropy; GFAP, glial fibrillary acidic protein; LC, locus coeruleus; MD, mean diffusivity; ODI, orientation dispersion index.

integrity metrics (e.g., axial and radial diffusivity)^{7–9} or with higher CSF- or PET-derived measures of tau burden.^{10,36} While previous studies examined later disease stages, our results include advanced biophysical diffusion modeling to demonstrate that plasma-derived p-tau pathology is related to early alterations in fiber coherence in asymptomatic individuals. This is compatible with underlying axonal and myelin degeneration processes in vulnerable regions during the preclinical phase and suggests that such pathological processes continue across AD stages. Importantly, the LC-EC tract axons are late and thinly myelinated, which could make them particularly vulnerable to premature axonal injury.²⁵ Finally, our findings align with a broader range of studies investigating WM integrity in other tracts affected both in preclinical³⁷ and AD cohorts^{7–9,38–42} and demonstrating the role of WM tract health in the deposition or propagation of tau.^{43,44} Given the pathway's potential involvement in early tau pathology propagation, our findings suggest the presence of initial WM alterations within a healthy cohort characterized by relatively low levels of amyloid burden.

We identified a positive relationship between ODI and p-tau burden in our cohort. Higher ODI values typically indicate loss of fiber coherence in WM, but recent studies indicated that its interpretation depends on local-, tract-, or population-dependent WM characteristics. Previous work showed higher ODI along with elevated diffusivity and lower anisotropy in MCI patients,⁴¹ whereas another study reported lower WM ODI values in MCI and AD compared to healthy controls, but no differences in FA in the MCI group.⁴⁵ This discrepancy may reflect selective sparing or degeneration of crossing fibers, where

degradation of secondary fibers within selected voxels would make the primary bundle fibers appear more aligned, thereby increasing axonal coherence and reducing ODI.⁴⁶ Our findings in the LC-EC tract indicate that higher ODI is detrimental, as evidenced by the positive relationship between ODI and p-tau burden. Hyperphosphorylation of tau contributes to cytoskeletal destabilization and neurite and axonal disarray.⁶ Neuropathological observations show the presence of early hyperphosphorylated tau in LC-related dendrites and long axons,⁶ and we speculate that these tau-related morphological changes can be detected as lower microstructural ODI-derived fiber organizational coherence in the LC-EC tract. Of particular interest, the observed positive association between p-tau231 and LC-EC ODI supports the proposition that advanced biophysical diffusion models combined with early biomarkers may capture earlier pathological processes disrupting the structural integrity of the LC-EC tract than standard DTI models.

Second, we observed that lower FA and higher MD in the LC-EC tract corresponded to higher plasma GFAP levels. The LC-NE system is implicated in the modulation of neuroinflammatory processes by downregulating the microglial functions and the transcription of pro-inflammatory genes in astrocytes.^{14,47–49} Conversely, destroying LC neurons with the selective noradrenergic neurotoxin N-(2chloroethyl)-N-ethyl-2-bromobenzylamine (DSP-4) led to NE depletion and exacerbated neuroinflammatory responses and astrocyte reactivity in transgenic mice.⁵⁰ Reactive astrocytic dysregulation is marked by the overexpression of GFAP, with both animal and human neuropathology studies demonstrating that elevated GFAP levels in aging⁵¹ may contribute to less clearance of A β ⁴⁹ and presence

of both neurofibrillary tangles and A β plaques.⁵² We now extend these observations by additionally demonstrating that higher GFAP is associated with lower microstructural properties of the LC-EC tract.

In a series of exploratory analyses probing the topographical specificity of the relationships between LC-EC tract integrity and plasma AD-related biomarkers, we found that associations with plasma markers mostly concentrated at the bending points of the tract and toward its terminal portion. Although speculative, the evolution of FA, MD, and ODI values across the LC-EC tract could indicate that local WM integrity disruption reflects local characteristics of sharply bending portions within the bundle, with the parts of the tract exhibiting greater curvature or tissue stiffness being more vulnerable to biomechanical stress.⁵³ Importantly, these loci display similar robustness in the diffusion measures, as confidence intervals for all metrics are consistent throughout the tract. Besides the expected fanning of fibers as they approach their cortical targets, the observation that WM bundle portions closer to the EC exhibit higher axonal injury compared to the initial, LC-efferent part of the pathway is consistent with the proposed resilience of the LC throughout the development of AD pathology⁵⁴ and the neurodegenerative molecular mechanisms determining the selective vulnerability of EC neurons during AD pathogenesis.⁵⁵ Furthermore, local differences in the distribution of 4-repeat (4R) and 3-repeat (3R) tau isoforms along the tract might contribute to a higher occurrence of EC-biased associations. The LC harbors pretangle tau inclusions for a prolonged phase during the disease, with pretangle neurons exhibiting 4R-selective immunoreactivity, while the EC might exhibit tau pathology with a higher proportion of 3R isoforms,^{56,57} relating differently to neuronal health. The specificity of our results is further corroborated by the control tract analysis, showing no significant associations between microstructural integrity metrics and tau pathology or astrogliosis. In contrast, WM degeneration in these (pre-)frontal regions was associated only with plasma amyloid burden, consistent with the topography of A β .^{58,59}

Our study has several strengths. First, previous studies focusing on the structural integrity of the LC-EC pathway in patients along the AD continuum⁷⁻⁹ used a normative atlas of the tract, and they either examined a clinical cohort^{7,8} or considered previously diagnosed preclinical stages, with no explicit information on the range in underlying pathological burden.⁹ Our study combined recently developed plasma AD markers and dedicated ultra-high-field LC imaging, allowing us to identify the LC and track the LC-EC pathway at the individual level with an improved resolution, and the incorporation of advanced NODDI modeling alongside traditional DTI modeling to facilitate the interpretation of¹³ different neurobiological processes.²¹ However, our study also has limitations. First, our biomarker quantification method lacks spatial specificity when compared to similar assessments using PET, preventing us from drawing conclusions about the tau topography in the cortex. However, plasma markers are likely more sensitive in earlier stages as they also capture soluble forms, which is particularly important when investigating the earliest tau-related changes to the LC. Second, despite the uniqueness of the ultra-high-field MRI data and dedicated LC sequences, our dataset was relatively small, resulting in several marginally significant results after the FDR cor-

rection. Therefore, these results need to be interpreted with caution and require replication in future studies using a larger sample size. Finally, our cross-sectional approach limits causal interpretation, and our research question would benefit from a longitudinal framework to better elucidate the temporal ordering of the events at play.

5 | CONCLUSION

In conclusion, we provide in vivo evidence that elevated tau burden and astrocyte reactivity are associated with lower structural integrity within the LC-EC tract, marked by axonal degeneration and fiber disorganization. To our knowledge, this study is among the first investigating the LC-EC tract integrity in a healthy cohort with availability of both ultra-high-field imaging data and multiple plasma AD-related biomarkers. Our results show that microstructural alterations of the LC-EC tract reveal insights into the earliest AD-related pathophysiological processes, which may hold promise in identifying individuals at risk of progressing along the AD trajectory.

ACKNOWLEDGMENTS

Yuliya Patsyuk (Conceptualization, Methodology, Formal analysis, Visualization, Writing – original draft, Writing – review and editing), Maxime Van Egroo (Conceptualization, Methodology, Supervision, Writing – original draft, Writing – review and editing), Elise Beckers (Methodology, Writing – review and editing), Elouise A. Koops (Methodology, Writing – review and editing), Nicholas J. Ashton (Methodology, Writing – review and editing), Shorena Janelidze (Methodology, Writing – review and editing), Kaj Blennow (Methodology, Writing – review and editing), Oskar Hansson (Methodology, Writing – review and editing), Henrik Zetterberg (Methodology, Writing – review and editing), Benedikt A. Poser (Writing – review and editing), and Heidi I. L. Jacobs (Conceptualization, Funding acquisition, Project administration, Supervision, Writing – original draft, Writing – review and editing). This work was supported by Alzheimer Nederland (WE.03-2019-02, Heidi I. L. Jacobs), Alzheimer's Association (AARG-22-920434, Heidi I. L. Jacobs) and the Bright-Focus Foundation (A20211016F, Maxime Van Egroo). Maxime Van Egroo is supported by European Union's Marie Skłodowska-Curie Actions (101109451-ADEEPSLEEP). Kaj Blennow is supported by the Swedish Research Council (2017-00915), the Alzheimer Drug Discovery Foundation (ADDF), USA (RDAPB-201809-2016615), the Swedish Alzheimer Foundation (AF-930351, AF-939721, and AF-968270), Hjärnfonden, Sweden (FO2017-0243 and ALZ2022-0006), the Swedish state under the agreement between the Swedish government and the County Councils, the ALF-agreement (ALFGBG-715986 and ALFGBG-965240), the European Union Joint Program for Neurodegenerative Disorders (JPND2019-466-236), the National Institute of Health (NIH), USA, (grant 1R01AG068398-01), and the Alzheimer's Association 2021 Zenith Award (ZEN-21-848495). Oskar Hansson is supported by the Swedish Research Council (2016-00906), the Knut and Alice Wallenberg foundation (2017-0383), the Marianne and Marcus Wallenberg Foundation (2015.0125), the Strategic

Research Area MultiPark (Multidisciplinary Research in Parkinson's disease) at Lund University, the Swedish Alzheimer Foundation (AF-939932), the Swedish Brain Foundation (FO2021-0293), Parkinson Foundation of Sweden (1280/20), the Cure Alzheimer's Fund, the Konung Gustaf V:s och Drottning Victorias Frimurarestiftelse, the Skåne University Hospital Foundation (2020-O000028), Regionalt Forskningsstöd (2020-0314), and the Swedish federal government under an ALF agreement (2018-Projekt0279). Henrik Zetterberg is a Wallenberg Scholar and a Distinguished Professor at the Swedish Research Council supported by grants from the Swedish Research Council (2023-00356; 2022-01018, and 2019-02397), the European Union's Horizon Europe Research and Innovation Programme under grant agreement no. 101053962, Swedish State Support for Clinical Research (ALFGBG-71320), the Alzheimer Drug Discovery Foundation (ADDF), USA (201809-2016862), the AD Strategic Fund and the Alzheimer's Association (ADSF-21-831376-C, ADSF-21-831381-C, ADSF-21-831377-C, and ADSF-24-1284328-C), the Bluefield Project, Cure Alzheimer's Fund, the Olav Thon Foundation, the Erling-Persson Family Foundation, Stiftelsen för Gamla Tjänarinnor, Hjärnfonden, Sweden (FO2022-0270), the European Union's Horizon 2020 Research and Innovation Programme under the Marie Skłodowska-Curie grant agreement no. 860197 (MIRIADE), the European Union Joint Programme—Neurodegenerative Disease Research (JPND2021-00694), the National Institute for Health and Care Research University College London Hospitals Biomedical Research Centre, and the UK Dementia Research Institute at UCL (UKDRI-1003). Elouise A. Koops is supported by an Alzheimer's Association Research Fellowship (23AARF-1026796). Benedikt A. Poser is supported by H2020 FET-Open AROMA grant agreement no. 88587. Heidi I. L. Jacobs is supported by NIH grants R01AG062559, R01AG06806, R01AG082006, and R21AG074220.

CONFLICT OF INTEREST STATEMENT

Elouise A. Koops is the program chair of the Neuromodulatory Subcortical Systems Professional Interest Area of ISTAART. Nicholas J. Ashton has given lectures in symposia sponsored by Eli Lilly and is an associate editor at Alzheimer's Research & Therapy. Kaj Blennow has served as a consultant, on advisory boards, or on data monitoring committees for Abcam, Axon, BioArctic, Biogen, JOMDD/Shimadzu, Julius Clinical, Lilly, MagQu, Novartis, Ono Pharma, Pharmatrophix, Prothena, Roche Diagnostics, and Siemens Healthineers, is a co-founder of Brain Biomarker Solutions in Gothenburg AB (BBS), which is a part of the GU Ventures Incubator Program (outside submitted work), and is a member of the editorial board at *Alzheimer's Research & Therapy*. Oskar Hansson is an employee of Lund University and Eli Lilly. Henrik Zetterberg has served on scientific advisory boards and/or as a consultant for AbbVie, Acumen, Alector, Alzinova, ALZpath, Amylyx, Annexon, Apellis, Artery Therapeutics, AZTherapies, Cognito Therapeutics, CogRx, Denali, Eisai, Enigma, LabCorp, Merck Sharp & Dohme, Merry Life, Nervgen, Novo Nordisk, Optoceutics, Passage Bio, Pinteon Therapeutics, Prothena, Quanterix, Red Abbey Labs, reMYND, Roche, Samumed, ScandiBio Therapeutics AB, Siemens Healthineers, Triplet Therapeutics, and Wave, has given lectures sponsored by Alzecure,

BioArctic, Biogen, Celectricon, Fujirebio, LabCorp, Lilly, Novo Nordisk, Oy Medix Biochemica AB, Roche, and WebMD, is a co-founder of Brain Biomarker Solutions in Gothenburg AB (BBS), which is a part of the GU Ventures Incubator Program, and is a shareholder of MicThera (outside submitted work). Heidi I. L. Jacobs is immediate past chair of the Neuromodulatory Subcortical Systems Professional Interest Area of ISTAART and served as advisory board member of ISTAART. These relationships are not related to the content of the manuscript. All other authors report no relevant conflicts. Author disclosures are available in the [Supporting Information](#).

CONSENT STATEMENT

The study was conducted in accordance with the Declaration of Helsinki, and the experimental protocol was approved by the local Medical Ethics Committee of the Maastricht University Medical Center. Written informed consent was obtained from all the participants prior to inclusion.

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